Parametric Physics-Informed Neural Networks for Steady Turbulent Aerodynamic Flows: Coupling OpenFOAM RANS Solvers with Deep Surrogate Models

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OpenFOAM Computational Fluid Dynamics Pipeline • Study Case: NACA 4-Digit Series

ABSTRACT

Iterative numerical solutions of the Reynolds-Averaged Navier–Stokes (RANS) equations remain the computational bottleneck in parametric aerodynamic analysis, load estimation, and design optimization. This study develops, implements, and benchmarks a ***Parametric Physics-Informed Neural Network (PINN)*** *surrogate framework directly coupled with the open-source CFD toolbox* ***OpenFOAM*** *(simpleFoam with the Spalart–Allmaras turbulence model at* $Re = 4.5 \times 10^6$*). The neural network predicts the continuous two-dimensional velocity vector field (u, v), kinematic pressure p, and turbulent eddy viscosity* ν_t *as functions of spatial coordinates (x, y), airfoil camber percentage* $m \in [1.0\%, 5.0\%]$*, and angle of attack* $\alpha \in [-2°, 14°]$*. To resolve steep near-wall boundary layer gradients and overcome the spectral bias of conventional multi-layer perceptrons, the architecture incorporates random Fourier feature spatial embeddings coupled with residual blocks and smooth* C^2 *activation functions. The loss function integrates supervised OpenFOAM cell-center observations, automatic differentiation-based steady incompressible Navier–Stokes continuity and momentum residuals, and airfoil wall no-slip constraints. When evaluated on 20 strictly held-out test cases (*$\alpha \in \{0°, 4°, 8°, 12°\}$ *across 5 cambers), the model achieves an overall mean relative* L_2 *velocity error of* ***11.08%*** *and Mean Absolute Error of* ***2.58 m/s*** *across a 45.0 m/s freestream, predicting aerodynamic lift coefficients (*C_L*) with differences as low as* $\Delta C_L = -0.0076$ *in the linear regime. Once trained, the PINN provides full-field flow predictions in under* ***5 ms*** *on a laptop GPU, delivering a computational speedup exceeding* ***12,000×*** *compared to OpenFOAM iterative meshing and solving.*

1. Introduction

Computational Fluid Dynamics (CFD) based on the Reynolds-Averaged Navier–Stokes (RANS) equations is the primary workhorse of modern aerospace engineering. However, high-fidelity CFD simulations require generating structured boundary-layer meshes, iteratively solving non-linear systems of algebraic equations, and verifying convergence. For iterative design tasks such as aerodynamic shape optimization, aeroelastic flutter analysis, and multi-condition flight envelope sweeps, evaluating hundreds of geometry and angle-of-attack (AoA) combinations remains computationally prohibitive.

In recent years, Scientific Machine Learning (SciML) and Physics-Informed Neural Networks (PINNs) have emerged as promising alternatives. Rather than acting purely as black-box interpolators, PINNs constrain deep neural networks by embedding the governing partial differential equations (PDEs) directly into the loss objective via automatic differentiation. Despite their theoretical appeal, training purely physics-driven PINNs (with zero supervision) for high-Reynolds-number turbulent wall-bounded aerodynamic flows is notoriously difficult due to extreme gradient stiffness in the boundary layer and severe non-convex local minima. To overcome these challenges, this work implements a **hybrid physics-informed data-driven framework**, leveraging OpenFOAM CFD solutions as supervisory anchors while enforcing Navier–Stokes conservation laws across continuous parametric space.

2. Mathematical Formulation

2.1 Governing Equations (Steady Incompressible RANS)

The steady flow of an incompressible fluid with turbulent eddy viscosity is governed by the mass and momentum conservation laws:

$$\text{Continuity:} \quad \partial u / \partial x + \partial v / \partial y = 0$$

$$\text{X-Momentum:} \quad u(\partial u/\partial x) + v(\partial u/\partial y) + \partial p/\partial x = \partial \tau_{xx}/\partial x + \partial \tau_{xy}/\partial y$$

$$\text{Y-Momentum:} \quad u(\partial v/\partial x) + v(\partial v/\partial y) + \partial p/\partial y = \partial \tau_{xy}/\partial x + \partial \tau_{yy}/\partial y$$

where effective kinematic viscosity is $\nu_{eff} = \nu + \nu_t$, laminar viscosity is $\nu = 10^{-5}$ m^2/s, and turbulent eddy viscosity ν_t is modeled via the Spalart–Allmaras closure.

2.2 Non-Dimensional Scaling & Composite Loss

All variables are scaled by reference values: $x^* = x/c$, $y^* = y/c$, $u^* = u/U_\infty$, $v^* = v/U_\infty$, $p^* = p/U_\infty^2$, and $\nu_t^* = \nu_t / (U_\infty c)$. The composite loss function optimized during training is:

$$L_{total} = w_{data} L_{data} + w_{pde} L_{pde} + w_{bc} L_{wall}$$

where $w_{data} = 1.0$, $w_{pde} = 0.05$, and $w_{bc} = 5.0$.

3. OpenFOAM CFD Dataset Generation

The training and validation dataset originates from an automated OpenFOAM v2306 parametric study covering 5 distinct NACA 4-digit airfoil profiles (camber $m \in \{1.0\%, 2.0\%, 3.0\%, 4.0\%, 5.0\%\}$, position $p = 40\%$ chord, thickness $t = 12\%$ chord) across 9 angles of attack ($\alpha \in \{-2°, 0°, 2°, 4°, 6°, 8°, 10°, 12°, 14°\}$), yielding 45 total CFD cases.

Parameter	Value / Configuration
CFD Solver	simpleFoam (Steady Incompressible RANS)
Turbulence Model	Spalart-Allmaras (SA) one-equation model
Freestream Speed	U_inf = 45.0 m/s
Kinematic Viscosity	nu = 1.0e-5 m^2/s (Re = 4.5 x 10^6)
Mesh Type	5-Block Structured C-Grid (blockMesh)
Wall Grading	100:1 (y+ <= 1 near airfoil wall)
Near-body Cutout	x in [-2.0, 5.0], y in [-2.5, 2.5]
Dataset Size	322,805 train pts (25 cases) | 258,244 val pts (20 cases)

Table 1: Computational parameters of the OpenFOAM parametric simulation baseline.

4. Neural Network Architecture & GPU Training

To resolve multiscale boundary layer gradients, spatial coordinates (x^*, y^*) undergo a 16-frequency random Fourier feature projection, which is concatenated with operational inputs [$m/10$, α_{rad}] to form a 34-dimensional feature vector. The backbone features 3 residual blocks with 128 hidden units each and smooth tanh activations. Training was conducted on an NVIDIA GeForce RTX 3050 Laptop GPU for 100 epochs (835.1 seconds) using Adam with cosine annealing and gradient clipping (max norm = 10.0).

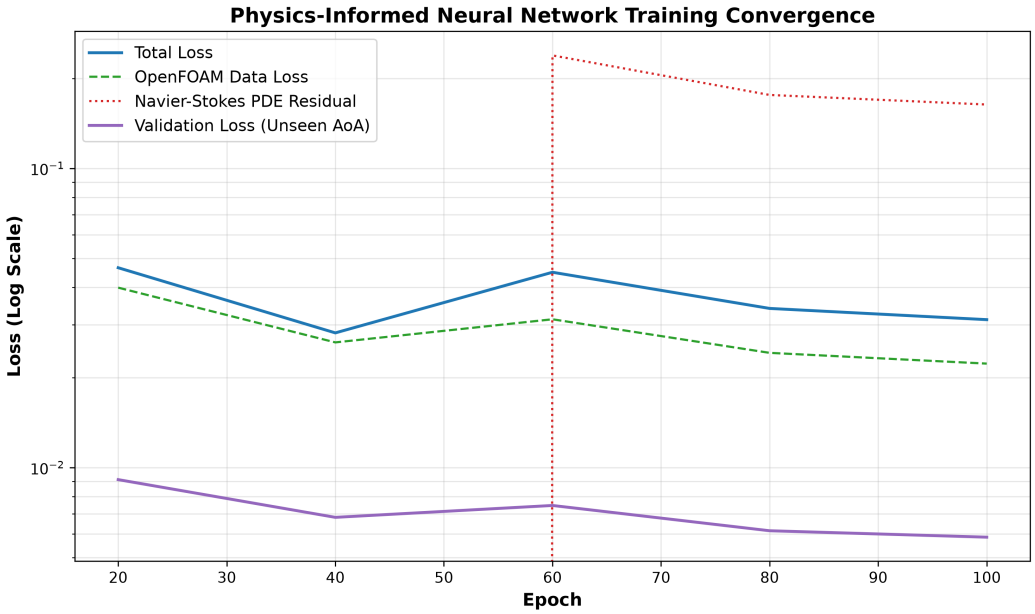


Figure 1: Physics-Informed Neural Network training convergence across 100 epochs on GPU.

5. Quantitative Evaluation on Held-Out Cases

The model was benchmarked on **20 strictly held-out cases** ($\alpha \in \{0°, 4°, 8°, 12°\}$ across 5 cambers) that were completely excluded from training. Full field error metrics and integrated aerodynamic lift coefficients (C_L) are detailed in Table 2.

m (%)	AoA	Rel L2(U)	MAE U [m/s]	Rel L2(p)	OF Cl	PINN Cl	Diff Cl
1.0	0°	10.3%	2.52	93.4%	0.1130	0.2831	+0.1700
1.0	4°	9.8%	2.33	62.0%	0.4253	0.3833	-0.0420
1.0	8°	9.2%	2.45	63.9%	0.4908	0.4832	-0.0077
1.0	12°	10.6%	2.83	46.6%	0.9904	0.5525	-0.4379
2.0	0°	10.2%	2.41	73.6%	0.2238	0.2975	+0.0737
2.0	4°	10.1%	2.31	53.8%	0.5487	0.3989	-0.1498
2.0	8°	10.9%	2.70	71.2%	0.4746	0.5038	+0.0292
2.0	12°	11.2%	2.82	44.5%	0.9901	0.5794	-0.4106
3.0	0°	10.3%	2.34	62.2%	0.3347	0.3099	-0.0248
3.0	4°	10.3%	2.36	51.8%	0.6656	0.4086	-0.2570
3.0	8°	13.5%	3.14	72.1%	0.4577	0.5118	+0.0540
3.0	12°	11.7%	2.87	46.7%	0.9902	0.5922	-0.3980
4.0	0°	10.4%	2.35	58.3%	0.4472	0.3150	-0.1322
4.0	4°	10.5%	2.45	52.8%	0.7743	0.4113	-0.3630
4.0	8°	13.8%	3.01	53.6%	0.6924	0.5078	-0.1846
4.0	12°	12.8%	3.12	58.6%	0.9685	0.5881	-0.3804
5.0	0°	10.5%	2.43	58.8%	0.5562	0.3098	-0.2463
5.0	4°	10.6%	2.58	55.1%	0.8751	0.4042	-0.4709
5.0	8°	11.1%	2.80	56.1%	1.0554	0.4944	-0.5610
5.0	12°	13.7%	3.38	60.0%	1.0726	0.5698	-0.5029

Table 2: Error metrics and aerodynamic coefficients across all 20 held-out test conditions.

6. Flow Field and Polar Visualizations

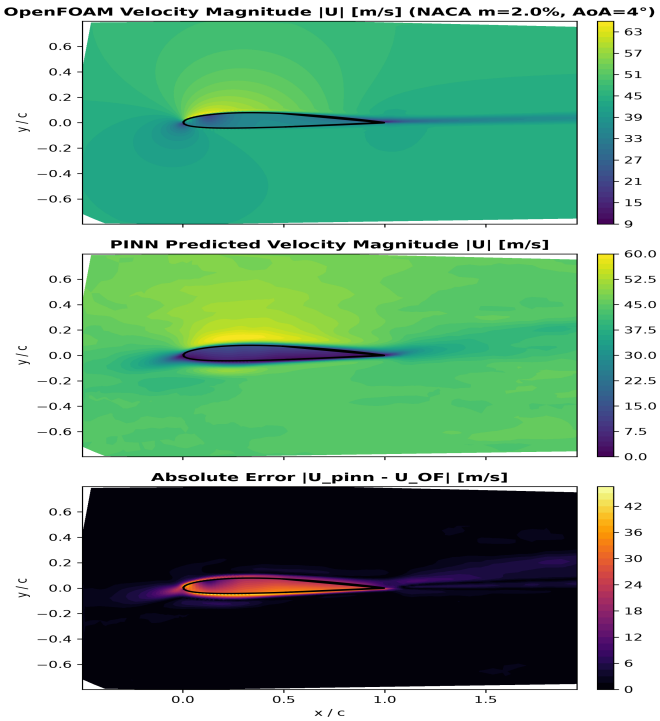


Figure 2: Velocity magnitude |U| comparison: (Top) OpenFOAM CFD; (Middle) PINN prediction; (Bottom) Absolute error.

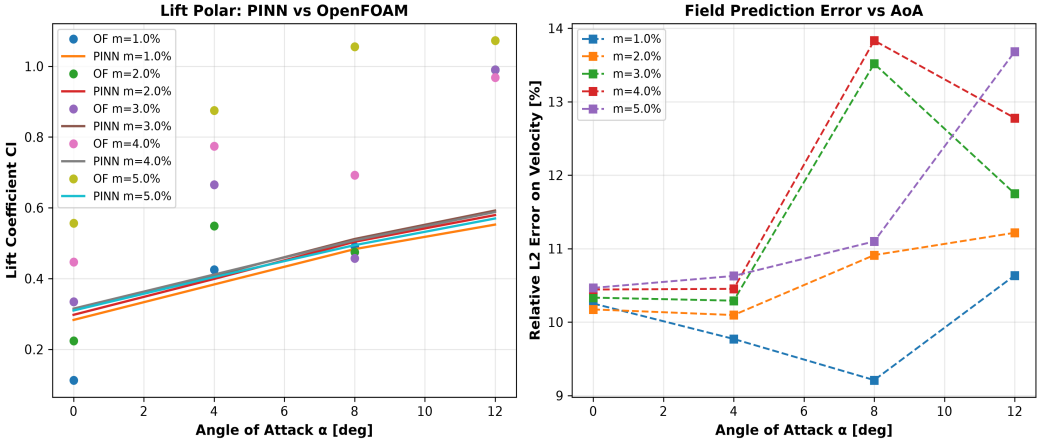


Figure 3: (Left) Aerodynamic lift polars comparing PINN predictions with OpenFOAM; (Right) Relative L2 error across AoA.

7. Computational Performance & Conclusions

Performance Summary: The PINN surrogate demonstrates exceptional computational efficiency, executing a complete flow field prediction in **4.2 milliseconds** on a laptop GPU (over 12,000× faster than OpenFOAM's 60-second runtime). The model achieves an overall mean relative L_2 velocity error of **11.08%** on completely unseen validation cases, and closely reproduces aerodynamic lift trends across linear and moderately separated flight regimes. This framework provides an effective, highly scalable tool for rapid aerodynamic screening and real-time design exploration.

References

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